Utility Billing Management Architecture

System Operational Specification & Functional Matrix Documentation

1. Executive Summary & Purpose

This architectural specification defines the operational criteria for the localized python-driven power utility management dashboard. The codebase provides real-time transactional metrics auditing, tier-based energy computations, and structural configurations to run clean user operational workflows.

2. Financial Tariff Slab Design

Consumer Category	Tier Slab Framework (kWh)	Base Rates ($ / kWh)	Fixed Access Fee
Domestic (D)	0 - 100 kWh 101 - 300 kWh > 300 kWh	$0.15 $0.25 $0.40	$5.00
Commercial (C)	0 - 200 kWh > 200 kWh	$0.30 $0.50	$15.00

3. System Code Integrity Constraints

- Input Validation Filters: Protect persistent modules from processing negative rollbacks or diminishing readings.
- Thread Persistence: Real-time serialization via standard vertical bar pipes configurations to isolate elements neatly.
- Data Scaling Protections: Strict conversion architectures prevent parsing unexpected alphanumeric inputs into numeric registers.